



12098 - AT2018cow: A Unique Laboratory for Super-Eddington Accretion and Black-hole Growth

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) SN-2018cow

ABSTRACT

Black-hole accretion via a disk plays a fundamental role in astronomy. In the past decade, a number of accretion-powered systems have been discovered that defy standard models, motivating new frameworks for disk evolution that accurately treat the poorly understood regime of super-Eddington accretion. An ideal test case for new disk-evolution models is AT2018cow, an accretion-powered transient in a galaxy only 60 Mpc away. We propose to observe AT2018cow's accretion disk using JWST and HST. We will model the newly observed UVOIR SED, together with existing

X-ray observations, using a new framework for the long-term evolution of accretion disks that takes super-Eddington accretion into account. The observations and modeling will enable us to both refine the modeling framework and infer the black-hole mass (a key degeneracy between competing progenitor models). JWST/NIRCam is essential because its angular resolution will enable AT2018cow to be separated from the background much better than the existing HST measurements. NIRCam's NIR sensitivity also has the potential to detect an underlying compact cluster, which would further help distinguish between progenitor scenarios and enable the cluster contribution to be jointly modeled with the accretion disk emission. A newly formed accretion disk in a very nearby system that underwent super-Eddington accretion is a rare opportunity: AT2018cow is a unique probe of a poorly understood regime of accretion physics that is likely critical for understanding massive BH growth at high redshift, as well as for determining the progenitors of new transient classes.

OBSERVING DESCRIPTION

We will use NIRCam to measure the SED of AT2018cow for a total of 16 hours. We will obtain imaging observations in five filters: F070W, F150W, F200W, F277W, and F356W. We will use NIRCam's simultaneous short- and long-wavelength observing capabilities, stacking observations to achieve the requisite sensitivity. Our readout strategy, SHALLOW4, minimizes the number of frames skipped during readout in order to maximize the likelihood of a detection. In addition, the number of integrations per exposure was optimized to ensure a >10 SNR for each short-wavelength bandpass. Stacked observations in F356W, as well as an observation in F277W, will yield > 5 -sigma detections. We avoid data excesses by only using one of the submodules (Module P) for imaging, rather than all NIRCam modules. The total time requested for our proposed observations is 16 hr and the data volume is 16.6 GB. To determine the appropriate observing strategy we used the JWST ETC.

NIRCam is the only instrument, on the ground or in space, which is capable of obtaining NIR observations at sufficient sensitivity and at sufficiently high angular resolution.

Our observations are designed to cover the transition to the Rayleigh-Jeans portion of the AT2018cow SED, and/or the peak of expected emission from a possible underlying compact cluster as predicted by some progenitor models. Observations any time during Cycle 5 are suitable.

To estimate the required S/N, we have made conservative assumptions: that there is no cluster contribution (so the emission is entirely Rayleigh-Jeans, decreasing in brightness as ν^2) and assuming the worst-case model parameters for fading disk emission (varying parameters including the disk viscosity). The emission is likely to be brighter than our assumptions. Even with some decrease in sensitivity due to uncertainty in modeling the background emission, we still have sufficient S/N for our modeling and interpretation.

JWST Proposal 12098 (Created: Friday, March 13, 2026, 5:06:52PM Eastern Standard Time) - Overview

We also request UV observations with HST WFC3/UVIS in order to have a near-simultaneous NIR through UV SED. We request F225W and F336W, the same filters as in the most recent AT2018cow observations (which were from January 2024). We assume standard overheads for guide-star acquisition (6 minutes), WFC3/UVIS overheads (2.6 minutes for first exposure, 2.1 thereafter), and dithering to provide adequate subpixel sampling of the point-spread function (20-30s). We request a single orbit for each filter, which will be sufficient to detect AT2018cow at very high significance (>10 sigma). There will be some reduction in significance due to uncertainty in modeling the background emission but with the long lever arm in wavelength between the NUV and NIR, the S/N will still be sufficient for our accretion-disk modeling.

Proposal 12098 - Targets - AT2018cow: A Unique Laboratory for Super-Eddington Accretion and Black-hole Growth

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SN-2018cow	RA: 16 16 0.2540 (244.0010583d) Dec: +22 16 5.10 (22.26808d) Equinox: J2000	Epoch of Position: 2000	
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Unidentified</i> <i>Description=[Ultraviolet sources]</i>					

Proposal 12098 - Observation 1 - AT2018cow: A Unique Laboratory for Super-Eddington Accretion and Black-hole Growth

Observation	Proposal 12098, Observation 1									Fri Mar 13 22:06:52 GMT 2026	
	Diagnostic Status: Warning										
	Observing Template: NIRCcam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
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Template	Module					Subarray					
	B					FULLP					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F070W	F356W	SHALLOW4	10	5	20	4	10693.823		
	2	F150W	F356W	SHALLOW4	10	5	20	4	10693.823		
	3	F200W	F356W	SHALLOW4	10	5	20	4	10693.823		
	4	F200W	F277W	SHALLOW4	10	6	24	4	12841.177		